NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

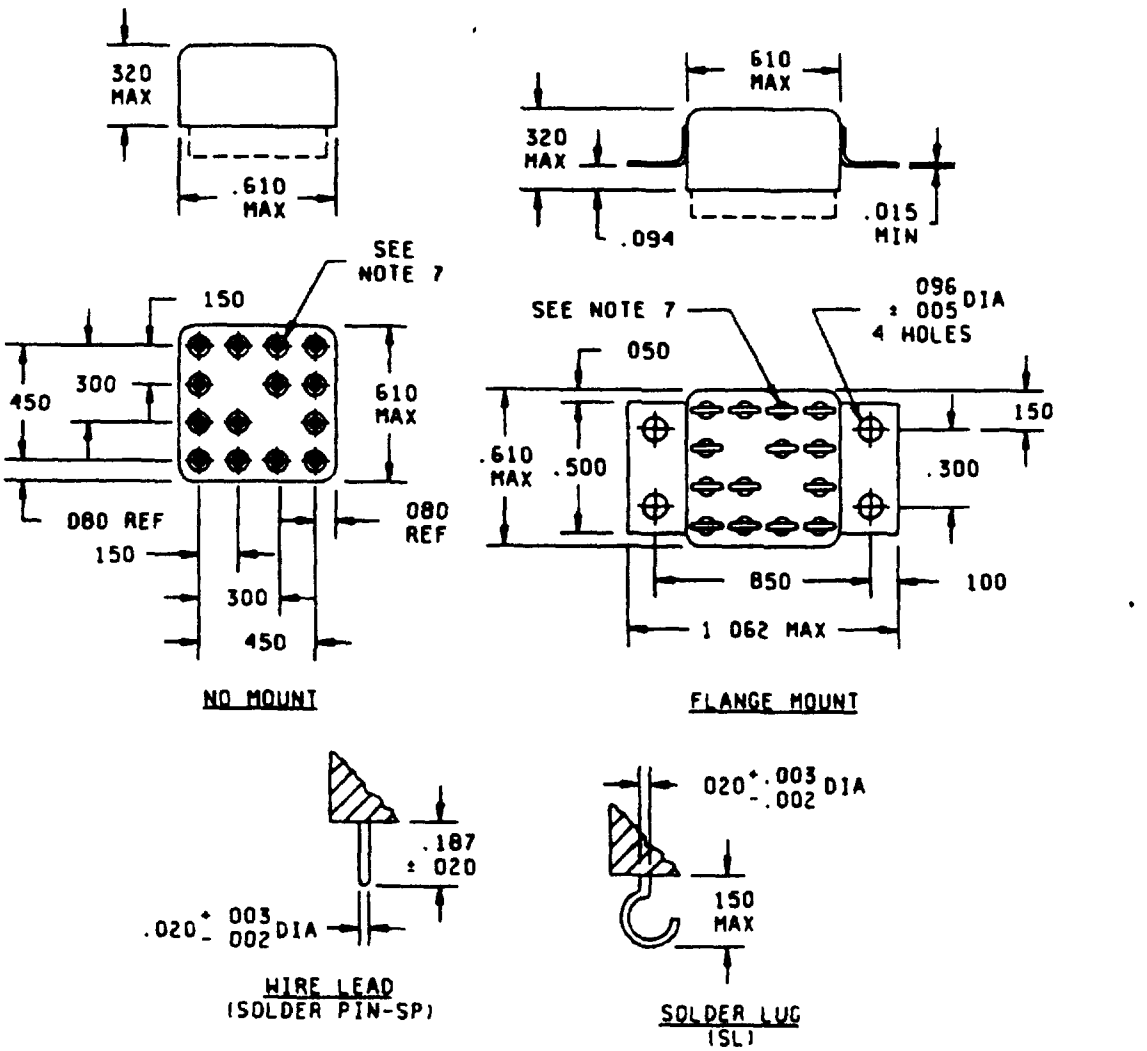
MIL PRF 39016/14D
20 JULY 1988
SUPERSEDING
MIL-R-39016/14C
13 March 1984

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT
LOW LEVEL TO 2 AMPERES (0 150-INCH TERMINAL SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense

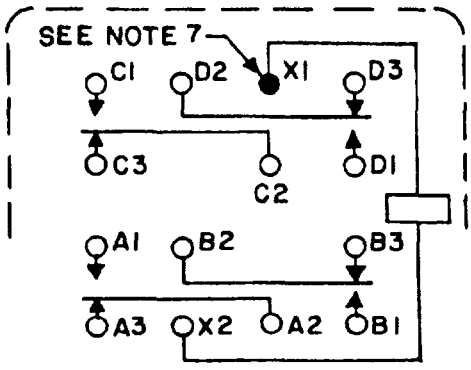
The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R39016



(1) FIGURE 1 Dimensions and configuration

(2) [illegible]

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CIRCUIT DIAGRAM
TERMINAL VIEW
DEENERGIZED POSITION

Inches	mm	Inches	mm
.001	0.03	.150	8.81
.003	0.08	.187	4.75
.005	0.13	.300	7.62
.020	0.51	.320	8.13
.050	1.27	.450	11.43
.080	2.03	.500	12.70
.094	2.39	.610	15.49
.096	2.44	.850	21.59
.100	2.54	1.062	26.97

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, the tolerance is ±.010 (0.25 mm).
4. Terminal locating dimensions shown are applicable to all type mounts.
5. The shape of solder lugs is optional.
6. Terminal numbers in circuit diagram are for reference only. Numbers do not appear on relay.
7. Indicated terminal shall be identified with a contrasting bead.
8. Coil symbol optional in accordance with MIL-STD-1285.

Ⓓ FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS.

CONTACT DATA.

Load ratings:

High level (relay case grounded).

(D) Resistive. 2 amperes at 28 V dc; 0.125 ampere at 115 V ac, (60 and 400 Hz) with case grounded. 0.5 ampere at 115 V ac, (60 and 400 Hz) with case not grounded. 400 Hz life test not required for qualification testing.

(D) Inductive. 0.5 ampere at 200 millihenries inductive at 28 V dc.

Lamp. 0.10 ampere at 28 V dc.

Low level 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current. Applicable.

Contact resistance or voltage drop.

(D) Initial. 0.050 ohm maximum.

High level.

During life. Not more than 5 percent of open circuit voltage.

(D) After life. 0.150 ohm maximum.

Low level.

During life: 33 ohms maximum.

(D) After life. 0.150 ohm maximum

(D) Intermediate current

During intermediate current. 1 ohm maximum.

After intermediate current. 3 ohm maximum

Contact bounce 2.0 milliseconds (ms) maximum. (Applicable to failure rate level "L")

Contact stabilization time: 2.5 milliseconds (ms) maximum (applicable to failure rate levels "M" "P", and "R").

Overload (high level only) 4 amperes resistive at 28 V dc. 1.0 ampere inductive at 28 V dc (ac not applicable).

COIL DATA. See table I.

Operate time 4.0 ms maximum over temperature range with rated coil voltage.

Release time. 4.0 ms maximum over temperature range from rated coil voltage

ELECTRICAL DATA.

Insulation resistance. 10,000 megohms minimum, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum

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Dielectric withstanding voltage

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure, and between all contacts in the energized and deenergized positions	750	350 All terminals to case
Between case, frame, or enclosure and coil(s)	500	
Between all contacts and coil(s)	750	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	750	

ENVIRONMENTAL DATA.

Temperature range -65°C to +125°C.

(D) Vibration (sinusodial) MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts

(D) Vibration (random). MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification and group C testing only

(D) Shock (specified pulse) MIL-STD-202, method 213, test condition B (75 g's) Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts

Magnetic interference· Applicable.

Resistance to soldering heat Applicable

Acceleration· Applicable

PHYSICAL

Terminals. See figure 1 and table I.

Terminal strength. 1.5 ±0.2 lbs (pull).

(D) Solderability: Applicable

Terminal twist test. Applicable to wire lead terminals

Dimensions and configuration: See figure 1.

Weight 8.5 grams (0.30 ounce) maximum.

Identification marking (full) Applicable.

LIFE TEST REQUIREMENTS.

High level· 100,000 cycles per relay.

(D) Low level· 100,000 cycles plus 900,000 cycles mechanical life.

PART NUMBER· M39016/14-(dash number from table I and suffix letter designating failure rate level)

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Ⓓ TABLE I. Dash numbers and characteristics. 1/

Dash number 2/	Mount	Leads	Coil voltage V dc 3/		At 25°C				Over temperature range		
			Rated	Max	Coil resistance Ohms ±10%	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
004	Flange	Lug	6	7	28	2.7	1.6	0.3	3.8	2.2	0.18
005	No mount	Wire (SP)									
009	Flange	Lug	9	11	73	4.2	2.5	0.4	5.6	3.4	0.25
010	No mount	Wire (SP)									
006	Flange	Lug	12	14	115	5.4	3.2	0.6	7.6	4.3	0.36
007	No mount	Wire (SP)									
011	Flange	Lug	18	22	280	8.4	5.0	0.8	11.2	6.7	0.5
012	No mount	Wire (SP)									
013	Flange	Lug	22	27	430	10.3	6.0	1.0	14.0	8.0	0.6
003	No mount	Wire (SP)									
001	Flange	Lug	26.5	35	720	13.5	8.1	1.5	18.0	10.8	0.9
008	Flange	Wire (SP)									
002	No mount	Wire (SP)									
014	Flange	Lug	36	42	1040	17.1	10.5	1.9	22.8	14.0	1.1
015	No mount	Wire (SP)									

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example, 001L - - - - - 003R.

3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay."

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(D) TABLE II. Qualification inspection and sample size.

Single submission	Group submission	
units plus 1 open unit for level L at C = 0 2/	M39016/14-001	units plus 1 open unit for level L at C = 0 2/
33 units plus 1 open unit for level M at C = 0 2/		33 units plus 1 open unit for level M at C = 0 2/
Qualification inspection as applicable.		Qualification inspection as applicable.
	M39016/14-004 M39016/14-006 M39016/14-009 M39016/14-011 M39016/14-014	2 units, each part number, qualification inspection table, group II.
	M39016/14-003	2 units, qualification inspection table, group II, and shock, vibration, acceleration, terminal strength, and seal.

1/ For retention of qualification or extension of qualification to low failure failure rate levels, all life test data accumulated on MIL-R-39016/53 and MIL-R-39016/54 may be used in addition to MIL-R-39016/14 data. Prior to retention of qualification testing, the relay manufacturer shall preselect the sampling plans.

2/ The number of units required for qualification testing will be increased as required in group IV, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to qualification testing, the relay manufacturer shall preselect the sampling plan.

Qualification inspection (reduced testing) and sample size. See table III. If the relays produced for MIL-R-39016/14 are similiar in construction and design except for the diodes and coils to the relays produced for MIL-R-39016/53 and /54, then reduced testing for qualification of MIL-R-39016/14 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/53 and /54

(D) TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage - Group II of the qualification inspection table. 1 unsealed sample unit - Internal examination.

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Supersession data See table IV

(D) TABLE IV Supersession data

New part no M39016/14- 1/	Superseded part no M5757/80-
004	016
006	017
001	018
005	025
007	026
002	027

1/ Complete part number shall contain suffix letter L, M, P or R to designate failure level rate (see 2/ of table I). A part with any failure rate supersedes the applicable MIL-R-5757 part

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
(D) Air Force - 85

Review activities
Army - AR
Navy - AS, OS, SH
(D) Air Force - 99
DLA - ES

User activities.
Navy - MC
(D) Air Force - 11, 19

Preparing activity
Navy - EC

Agent
DLA - ES

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